

Data Science

Lecture 01: Introduction

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Converted from LNKB-tex part-en-ds-chp1-introduction.tex. Examples and framing updated from the 2022 "big data" era to the current one the argumentative spine (data does not speak for itself) is unchanged and is what the rest of the course rests on. No prerequisites.

Module 1

The Data Deluge

Guiding Question: *We generate more data than we can store. Does having more of it make us know more?*

How Much Data Exists?

Year	Global datasphere	Per person on earth
2002	5 exabytes of new information	—
2010	~1 zettabyte	a trillion gigabytes in total
2020	~40 zettabytes	≈ 5,200 GB
2025	~175 zettabytes	≈ 21,000 GB

In 2007 the amount of information produced first **surpassed available storage capacity**. Since then, most data that is generated is never stored.

The exact figures matter less than the two structural facts: growth is exponential, and generation outruns storage — so *selection* is unavoidable and someone is choosing. Ask who decides what gets kept that is a data-science question before it is an engineering one.

Growth Is Not the Interesting Part

Volume

More bytes than anyone can read, index or curate.

Heterogeneity

Text, sensors, images, logs, transactions — rarely in a common schema.

Velocity

Streams that must be acted on before they can be stored.

The scarce resource is not data. It is **the ability to turn data into justified conclusions** — which is what this course is about.

Deliberately deflate the "big data" framing (P2): volume is a logistics problem. The course's subject is the inferential problem, which does not get easier with more rows — and sometimes gets harder, as the next unit shows.

Science at Scale

Large Hadron Collider

1 MB per event, 600 million events per second — roughly **18 zettabytes per year** if everything were kept.

A single tier-1 centre stores ~2.2 petabytes per year.

The filter is the experiment

Almost all events are **discarded in hardware**, in real time, by trigger rules written in advance.

What is kept is decided by a theory of what matters.

The LHC is the sharpest possible illustration that data is never raw: the trigger encodes physics theory, so the stored dataset is already theory-laden. This directly sets up the "end of theory" argument next.

Data at Web Scale

Search

Web-scale indexing processes petabytes per day and answers in milliseconds.

Platforms

Interaction logs — clicks, dwell time, scrolls — dwarf the content itself.

Models

Training corpora for language models are measured in trillions of tokens.

In every case the value comes from **aggregation across many weak signals**, not from any single record.

Data is never raw: at the LHC a physics theory decides what is recorded, and on the web value comes from aggregating signals that are individually worthless. **Someone always chose what the dataset contains.**

Speaker notes

Ask what a single click is worth versus a billion clicks — the answer motivates statistics as the discipline of extracting signal from individually uninformative observations. The takeaway is the bridge to Module 2.



Module 2

Does Data Speak for Itself?

Guiding Question: *Twenty-nine teams analysed the same dataset and the same question. Why did they not get the same answer?*

"The End of Theory"

With enough data, the numbers speak for themselves. Correlation supersedes causation, and science can advance without coherent models or unified theories.

Chris Anderson, *Wired*, 2008. The argument has returned, almost verbatim, in claims that sufficiently large models make domain theory unnecessary.

Is it right?

Guiding question of the module (P1). Do not answer yet. Ask for a show of hands and hold both camps — the next four slides supply the evidence, and students should feel the argument's pull before it is dismantled.



One Dataset, Many Answers

Twenty-nine research teams received **the same dataset** and the same question:

Are football referees more likely to give red cards to dark-skinned players than to light-skinned players?

The result

Odds ratios ranged from **0.89 to 2.93**.
Twenty teams found a significant effect
nine did not.

The reason

No team made an error. They made
different defensible modelling choices
— which covariates, which exclusions,
which model family.

The data did not determine the answer. The **analysis** did. (Silberzahn et al. 2018)

Speaker notes

The most important slide of the lecture. Emphasise that this is not a story about incompetence or fraud — it is what "researcher degrees of freedom" means. Ask what the honest reporting practice would be (pre-registration publishing the specification curve), and note that this is exactly what makes P3 (epistemic modelling) a professional obligation rather than a philosophical nicety.



Correlation Is Not Causation

Spurious correlations are easy to find and get **easier as datasets grow**: with enough variables, some pair will track another closely by chance alone.

Why it happens

Testing many pairs makes an extreme correlation likely *somewhere*, even when every pair is independent.

Why more data does not fix it

More rows shrink the confidence interval around a **meaningless** number. The estimate gets more precise, not more valid.

Speaker notes

The second box is the deep point and the one students miss: precision and validity are different axes. This returns in hypothesis testing, where a large n makes trivial effects "significant".



Summary Statistics Hide the Data

Datasets whose **summary numbers agree exactly** — same averages, same spread, same strength of association — can look completely different when plotted: a dinosaur, a star, a set of lines (Matejka & Fitzmaurice 2017).

We will make "average", "spread" and "association" precise in Lecture 02, and meet this phenomenon again there as Anscombe's quartet.

Plot the data. Then plot it again, differently.

Forward *pointer*, not a forward dependency (P7): the phenomenon is shown visually and named in plain words; the statistics that make it precise are Lecture 02's job, where Anscombe's quartet returns as consolidation. Do not use the words variance or correlation as if defined here.

Where This Leaves Us

Data does not speak for itself

Every dataset was selected, filtered and encoded by someone.

Data cannot be read intuitively

Human intuition fails at scale, in high dimensions, and with base rates.

So theory is needed *more*

Models make the assumptions explicit enough to be criticised.

Big data does not end theory. It **raises the cost of implicit theory**, because assumptions that stay unstated now get automated and scaled.

Return to the guiding question and answer it explicitly. The last sentence is the one to leave on the board: the danger is not wrong theory, it is invisible theory. This is the thread the whole course pulls on.



Module 3

What Is Data Science?

Guiding Question: *If data science is not statistics, not machine learning and not engineering, what is left that is its own?*

A Definition

Data Science researches and applies technologies, systems and algorithms to gain knowledge from potentially large, heterogeneous and dynamically changing data sets — for a particular domain. (Dhar 2013)

The phrase doing the work is "**for a particular domain**": without it, there is no criterion for which patterns count as knowledge.

Math-first equivalent for a definitional slide (P5): show it, then interrogate it. Ask which word could be deleted without loss (probably "large") and which could not ("domain", "knowledge").

Neighbouring Fields

Machine Learning

Learning from past experience to improve at a task in the future.

Artificial Intelligence

Building systems that behave intelligently.

Data Engineering

The twin field: managing, organising and providing access to data.

Data science is not a subset of any of them. It is the discipline of getting from a **domain question** to a **defensible answer**, using whichever of these tools fits.

Students arrive with these words used interchangeably. The distinction that sticks: ML is a method, data engineering is an infrastructure, data science is an inquiry.



Three Kinds of Task

Explorative

Find patterns.

Which customers behave similarly?

Descriptive

Summarise what is there.

What does an average customer spend?

Predictive

Anticipate future outcomes.

Is this transaction fraudulent?

Naming the task first constrains everything else — the method, the evaluation measure, and what counts as success.

Data science is defined by its **question and its domain**, not by its tools. Explorative, descriptive and predictive tasks demand different methods and different evidence — so the first decision in any project is which one you are doing.

This triple is the organising map of the whole course: clustering and projections are explorative, statistics is descriptive, classification and regression are predictive. Point forward to where each returns.



Module 4

How Knowledge Is Acquired

Guiding Question: A pattern in data is not yet knowledge. What has to happen in between?

Three Modes of Reasoning

	From	To	What it guarantees
Deduction <i>(logic)</i>	premises	what must follow	certain — but adds no new content
Induction <i>(statistics)</i>	observed patterns	a regularity , a hypothesis	nothing — content-adding, not guaranteed
Abduction <i>(probability)</i>	observations	the best available explanation	nothing — only comparable against rivals

Machine learning lives in the bottom two rows.



Three modes, one criterion to compare them: what does each guarantee, and what does it buy? Deduction guarantees everything and discovers nothing induction and abduction discover and guarantee nothing. Data science lives in the latter two, which is why validation is a permanent obligation.

The Scientific Reasoning Cycle

The three modes are not alternatives — they take turns.

Example: gravity.

Deduction

From the theory: all objects of equal mass fall at the same rate everywhere on earth.

Induction

Design an experiment, measure fall times at several locations, compute means and standard deviations.

Abduction

Times agree at one location but vary between locations.
Best explanation: distance from earth's mass centre.

The cycle turns: the abduced explanation becomes the next theory to deduce from. A data-science project runs the same loop, usually several times.

Speaker notes

Have students place a machine-learning project in the cycle: choosing a model family is deduction from assumptions, fitting is induction, and diagnosing why it fails on a subgroup is abduction. The loop is the honest picture of the workflow, unlike the tidy pipeline diagrams.



Mini-Exercise

A dashboard shows that customers who use the mobile app churn **less** than those who do not.

- 1 What would a purely **inductive** reading conclude?
- 2 Give two different **abductive** explanations.
- 3 What **deduction** would let you distinguish them?

Speaker notes

Question and solution share a slide: the three prompts stay visible while the answers appear, so students can check each answer against the question that produced it. One click reveals all three.

The exercise rehearses the whole cycle in a business setting and previews the correlation/causation problem. The key move is 3: a hypothesis earns its keep by predicting something checkable that its rival does not.



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INDUCTION

"App use reduces churn" — a regularity read straight off the data.

ABDUCTION

The app causes loyalty — *or* loyal customers install the app. Same correlation, opposite causal direction.

DEDUCTION

If the app causes loyalty, randomly prompting a subgroup to install it should lower their churn — an experiment deduced from the hypothesis.

What Counts as Knowledge?

Knowledge discovery in databases (KDD) is the non-trivial process of identifying patterns in data that are

- **valid** — they hold beyond the sample,
- **novel** — they were not already known,
- **relevant** — they matter for a decision,
- **interpretable** — they can be understood and acted upon.

A pattern that fails **any** of the four is not knowledge. Most patterns a large dataset offers fail at least one.

≡ Knowledge is a pattern that is **valid, novel, relevant and interpretable**. Data mining produces patterns only the four criteria turn one into knowledge — and most patterns fail at

least one of them.

Use the four criteria as a checklist against a real finding — say, "customers who buy nappies buy beer". Valid? Maybe. Novel? Perhaps. Relevant? Only if it changes a decision. Interpretable? Only with a mechanism. This is the antidote to pattern-hunting.

Module 5

The Process and the Craft

Guiding Question: *Most of a data-science project is spent before any model is fitted. On what, and why does it matter?*

Step 1 — Selection of Data

Types of source

- **Unstructured:** flat files, documents, text and multimedia
- **Structured:** relational databases, XML, web APIs

Types of data

media · object-relational · spatial and spatio-temporal · graphs and networks · time series and sequences · data streams

Selection is where the **first and largest** bias enters. What was not selected cannot be discovered later.

Tie back to the LHC trigger: selection encodes a theory of relevance. Ask what is systematically missing from a dataset of app users (everyone who never installed it) — survivorship bias, before it has a name.

Step 2 — Preprocessing

Selected

manually or
automatically, e.g. by
crawling

Cleaned

errors and noise
removed — *garbage
in, garbage out*

Integrated

combined with other
sources, resolving
conflicts

Transformed

into a reusable
representation for
analysis

This step routinely consumes most of a project's effort — and every choice in it is a modelling decision, not a clerical one.

Speaker notes

The "80% of the work" folklore is true but usually told as a complaint. Reframe it: preprocessing is where the representation α from the ML chapter is built, and the representation determines what can be learned at all.



Steps 3–5 — Mining, Evaluation, Deployment

Data mining

Apply the algorithm — clustering, classification, rule mining — that fits the task.

Evaluation

Test the patterns against the four criteria: valid, novel, relevant, interpretable.

Deployment

Put the result where a decision is made, and monitor it there.

The process is a **loop**, not a pipeline. Evaluation routinely sends you back to selection.

Speaker notes

The linear diagram is the standard depiction and the standard lie. Ask where the arrows back go — and note that in production, deployment feeds new data into selection, which is how feedback loops (and their pathologies) arise.



What the Work Actually Looks Like

Ask

Turn a domain question into something a dataset could answer — and check that it is the question worth answering.

Build

Acquire, clean, model, evaluate. Iterate, mostly backwards.

Communicate

Deliver a result someone can act on, with its uncertainty attached.

Speaker notes

The first and last boxes are the ones curricula under-serve and employers complain about. Say plainly that the middle box is what this course teaches, and the other two are where the value is realised or lost.



Three Skill Areas

Domain knowledge

What the data means, which values are impossible, which differences matter.

Statistics and modelling

What may be inferred, with what uncertainty, under which assumptions.

Engineering

Getting the data, at scale, reproducibly, into production.

The interesting failures happen **between** the three: a technically flawless model of the wrong quantity, or a correct analysis nobody can deploy.

The process is a **loop, not a pipeline**, and it spans three skill areas that nobody holds equally. The characteristic failures happen at the seams — a flawless model of the wrong quantity, or a correct analysis nobody can deploy.

Resist the "unicorn" framing — nobody holds all three deeply. The realistic skill is knowing enough of the neighbouring areas to ask the right question of a colleague, which is also why the course insists on stating assumptions aloud.

Module 6

Applications, and Their Failures

Guiding Question: What happens to a model's validity once the world can see the model?

Web and Social Media

Google Flu Trends

Search queries were used to nowcast influenza faster than public health reporting.

And then it failed

It overestimated cases substantially, because search behaviour changed — partly *because of media coverage of the tool itself*.

A model that acts on the world **changes the world it measures**. Validity is not a property a model keeps once earned.

Speaker notes

The failure is more instructive than the success and belongs on the slide. It is the cleanest example of a feedback loop and previews model monitoring, distribution shift, and the KDD loop from the previous unit.



Text and Language

Classic tasks

Summarisation, entity extraction, sentiment analysis, question answering — each once a research programme of its own.

Today

A single pretrained model handles all of them, with the task specified in words rather than in code.

What did **not** change: the need to evaluate on data the model has not seen, and to state what the output may be used for.

Speaker notes

The students' baseline experience is post-LLM, so the historical framing lands better as "these were separate fields" than as a list of tasks. The note is the continuity claim the course rests on.



Business and the Public Sector

Reported potentials from the era of the first big-data wave: large retail revenue gains, hundreds of billions in healthcare savings, comparable figures for e-government.

These are **projections from consultancy reports**, not measured outcomes. Treat them as claims to be interrogated — whose baseline, whose costs, and what happened afterwards?

Applications show what is possible their failures show what must be checked. Ask of every deployed model: what changed in the world because it exists?

Speaker notes

Explicit critique practice (P2 and the AI-context section): give students a confident number and ask what evidence would be needed to believe it. Ending the lecture on scepticism sets the tone for the rest of the course.



Image Credits & References

Dhar, Vasant (2013). Data science and prediction. *Communications of the ACM*.

Matejka, Justin and Fitzmaurice, George (2017). Same stats, different graphs: generating datasets with varied appearance and identical statistics through simulated annealing. *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*.

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